Product Categorization Using Machine Learning and Lightweight Embeddings for E-Commerce

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***Abstract*—The exponential growth of e-commerce platforms has intensified the need for automated, scalable, and accurate product categorization systems. This paper presents a machine learning framework for multi-class classification of product listings on Etsy using structured metadata and textual descriptions. We evaluate a diverse set of modeling approaches, including traditional machine learning methods such as Logistic Regression, Multinomial Naive Bayes, XGBoost, and LinearSVC, alongside pretrained and fine-tuned sentence embeddings. Our final ensemble model combining TF-IDF based XGBoost and Logistic Regression with a MiniLM-based MLP achieves a weighted F1 score of 0.89 on the top category prediction task. For bottom category prediction, LinearSVC trained on TF-IDF features achieves a weighted F1 score of 0.62, outperforming baseline models and lighter embedding approaches. Additional contributions include embedding visualizations using t-SNE and a comparative performance analysis of pretrained versus fine-tuned representations. The findings demonstrate that hybrid architectures integrating structured metadata and contextual embeddings yield robust performance while maintaining scalability for real-world e-commerce deployments.**

***Index Terms*—product categorization, e-commerce, machine learning, semantic embeddings, ensemble learning**

I. Introduction

In recent years, the exponential growth of e-commerce platforms has led to an increasing demand for automated, scalable, and accurate product classification systems. Platforms such as Etsy, which host millions of unique handmade and vintage items, rely on effective categorization to improve search relevance, enhance user experience, and streamline backend inventory management. However, due to the diversity and subjectivity of user-generated product metadata, traditional rule-based classification approaches often fall short in capturing the nuanced semantics and varied categorical hierarchies found in such marketplaces.

This paper focuses on developing a machine learning-based solution for predicting product categories using Etsy listing metadata. Specifically, the task involves assigning each product listing to its corresponding top and bottom category ID based on available textual and categorical attributes such as title, description, tags, material, color, and other handcrafted metadata fields.

The main objective of this research is to leverage the training dataset to learn patterns and accurately predict both top and bottom category IDs for unseen test data. The goal is to maximize the F1 score for each class across these two attributes. We evaluate both classical and modern machine learning approaches, including TF-IDF-based logistic regression, XGBoost, and ensemble models, alongside transformer-based methods such as MiniLM sentence embeddings, to assess their effectiveness in modeling user-generated e-commerce metadata.

Recent studies underscore the practical effectiveness of lightweight NLP pipelines for product categorization in e-commerce. Reddy et al. [1] demonstrate how TF-IDF vectorization, combined with logistic regression and XGBoost, can produce strong baseline results in multi-class classification settings involving short and noisy product texts. Their work also highlights the importance of leveraging attribute-specific tokenization to reduce sparsity and improve class separability in models trained on handcrafted product metadata. In parallel, Alkhalil et al. [2] take a more sophisticated approach, employing transformer-based architectures such as BERT for trust factor extraction from reviews and listings. They propose a segmentation-aware strategy to enhance contextual modeling, showing that incorporating token-level factor extraction improves the semantic granularity of the learned representations.

By synthesizing insights from both traditional and deep learning-based approaches, this study aims to explore a balanced solution that can be applied to real-world e-commerce taxonomies.Through systematic experimentation, we aim to derive insights into the strengths and limitations of different modeling strategies for real-world multi-class product categorization.

II. Related Work

Recent advances in natural language processing (NLP) and machine learning (ML) have significantly improved the automation of product classification and sentiment understanding

in e-commerce. Early work in this space applied traditional classifiers on vectorized textual inputs, but modern methods increasingly rely on deep models, ensemble techniques, and refined feature engineering strategies to boost performance across tasks like product categorization, review classification, and consumer sentiment analysis.

A foundational effort by Shalu et al. [3] focused on sentiment analysis for Apple product reviews using a combination of NLP preprocessing and classical classifiers, demonstrating that even standard models like NB and Logistic Regression (LR) can be effective when paired with appropriate features. In a similar direction, Kumar et al. [4] performed an in-depth comparison of transfer learning via fine-tuned BERT against conventional models. They showed that fine-tuning BERT significantly boosts classification accuracy across diverse domains, underscoring the importance of domain adaptation. This complements the findings of Shao et al. [5], who leveraged big data analytics and NLP to improve product classification. Their architecture integrated unstructured textual data with structured product features and demonstrated enhanced performance with hybrid approaches.

A detailed comparative evaluation by Xiao et al. [6] between BERT and traditional ML models in review classification revealed that while transformer-based models yield superior accuracy, classical models like SVM and NB remain competitive with appropriate feature engineering. The use of domain-specific corpora was identified as a crucial factor influencing BERT's performance.

Broader surveys provide a macro perspective. Xue Zhang et al. [7] summarized ML and DL techniques applied in e-commerce, noting trends such as the rise of recommendation systems, fraud detection, and sentiment modeling using RNNs, GANs, and BERT. They emphasized the challenge of data sparsity and the necessity of personalization. Meanwhile, M. Zhang [8] demonstrated the utility of LSTM networks for sentiment classification, highlighting their effectiveness in modeling sequential dependencies in user-generated content.

Jayadharshini et al. [9] presented a comprehensive classification framework using multiple models including Linear SVC, RF, and NB. Their experimental results showed Linear SVC outperforming others on product classification tasks, especially when paired with bigram and trigram TF-IDF features. The strength of this work lies in its detailed preprocessing strategy and feature engineering depth. Complementary to task-specific models, Taha et al. [10] provided a broad survey of text classification approaches, categorizing them based on methodology (e.g., shallow vs. deep learning) and application domains. The review highlighted the importance of feature representation, classifier choice, and dataset characteristics in achieving optimal results. Gasparetto et al. [11] offered further insights by delineating the data flow from raw text to predictions, emphasizing the role of preprocessing, especially in multilabel settings.

Recent additions offer useful augmentations. Rani et al. [12] investigated multiple vectorization techniques, BoW, TF-IDF, and N-gram models, comparing their performance across classification algorithms. They concluded that hybrid vectorization methods generally yield better performance due to increased contextual information. Aydoğan and Okay [13] extended this analysis by experimenting with five ML algorithms across various vectorization schemes. Their findings affirm the superiority of ensemble methods like RF and XGBoost, especially when using TF-IDF and CountVectorizer with balanced sentiment classes. They also identified a performance drop when employing N-grams, likely due to overfitting and increased feature sparsity.

Together, these studies underline the evolving landscape of text-based product classification and sentiment analysis in e-commerce. There is a clear trajectory from basic bag-of-words models to deep contextual embeddings, from rule-based pipelines to optimized ML workflows. Effective preprocessing, hybrid feature engineering, and model tuning remain essential for achieving state-of-the-art performance, particularly when dealing with high-dimensional, domain-specific textual data.

III. Methodolody

A. *Dataset*

This study utilizes a dataset of over 229,000 product listings from an e-commerce platform known for handcrafted and vintage items. Each listing includes textual fields such as the title, description, and tags, along with structured metadata like product type, room or usage context, recipient, material, occasion, and color. The items are annotated with two levels of categorical labels: a broad top-level category and a more granular bottom-level category.

B. *Data Preprocessing*

To prepare the data for modeling, the title, description, and tags were merged into a single text input, enabling a unified representation of each product. The combined text was standardized by converting to lowercase, removing special characters, and stripping HTML artifacts and extra spaces.

Two main strategies were employed to convert the cleaned text into numerical features. The first was TF-IDF vectorization using unigrams, bigrams, and trigrams to retain both word importance and local context. The second utilized pretrained transformer models to obtain dense sentence embeddings that capture deeper semantic meaning.

Categorical features with missing values were imputed with a placeholder and one-hot encoded for compatibility with machine learning models. This preprocessing pipeline ensured the dataset was clean, consistent, and suitable for training a wide range of models, from linear classifiers to neural architectures.

C. *Top Category Classification*

The top category classification task was framed as a multi-class problem. Given the variability in product descriptions, the goal was to develop a solution that was not only accurate but also robust and generalizable.

Initial experiments employed TF-IDF vectorization to extract frequency-based features from the cleaned product text.

Multiple linear models were evaluated on this representation, including logistic regression, support vector machines, ridge classifiers, and SGD classifiers, all known for their effectiveness on high-dimensional data.

To capture richer semantic context, dense sentence embeddings were generated using the MiniLM transformer model. These embeddings served as input to a multilayer perceptron (MLP), enabling the model to capture deeper linguistic nuances beyond what TF-IDF could represent.

After evaluating individual models, a weighted ensemble was built by combining predictions from three classifiers: logistic regression and XGBoost (both trained on TF-IDF features), and an MLP trained on MiniLM embeddings. Each model contributed unique strengths, XGBoost added non-linear capabilities, logistic regression provided a strong linear baseline, and MiniLM enabled semantic understanding. The final predictions were generated using a soft voting mechanism with tuned weights, slightly favoring XGBoost. Fig. 1 illustrates the architecture of this ensemble model.

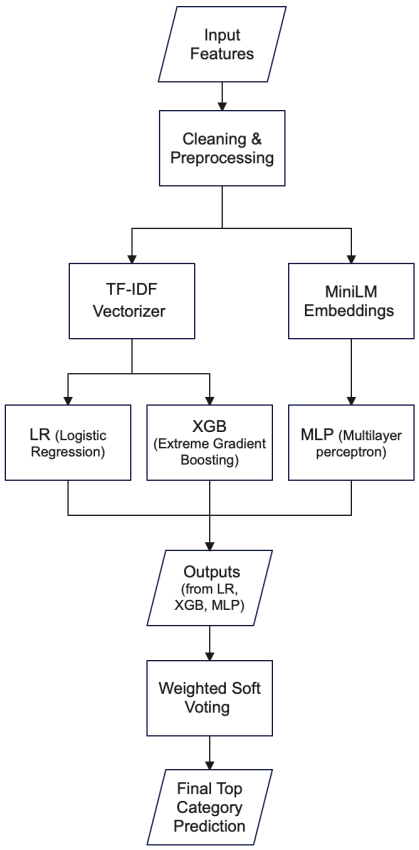

Fig. 1. Architecture of the final ensemble model used for top category classification.

D. *Bottom Category Classification*

Bottom-level category prediction posed a greater challenge than top-level classification due to a significantly larger label space, higher class imbalance, and increased semantic overlap between categories.

The modeling process began with TF-IDF representations combined with simple classifiers such as Multinomial Naive Bayes and logistic regression. To handle large-scale, sparse data more efficiently, stochastic gradient descent (SGD) classifiers were introduced next. These allowed for batch-wise training and offered a balance between scalability and performance.However, further testing revealed that LinearSVC consistently outperformed SGD-based models, particularly in terms of stability and handling imbalanced classes.

In parallel, semantic embeddings from the MiniLM transformer model were also explored. These were used as inputs to neural networks like MLPs and in hybrid models alongside categorical metadata. Oversampling techniques were applied carefully to rebalance rare categories and avoid overfitting.

Despite these efforts, more complex or resource-intensive architectures did not surpass the performance of the simpler TF-IDF + LinearSVC setup. Fig. 2 illustrates the structure of the final model.

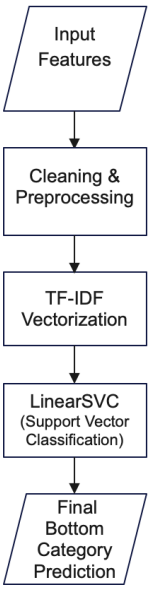

Fig. 2. Workflow of the LinearSVC-based model used for bottom category classification.

IV. EVALUATION

The performance of all models was primarily evaluated using the weighted F1 score, which considers both precision and recall while accounting for class imbalances. This metric was chosen due to the inherent skew in category distribution across the dataset, where some product classes are significantly more frequent than others. To ensure consistency and fairness in comparison, the same training and validation splits were used across all models.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{1}$$

$$\text{Weighted F1-Score} = \sum_{i=1}^{K} \frac{n_i}{N} \cdot \text{F1}_i \tag{2}$$

where K is the total number of classes, n_i is the number of true instances for class i, N is the total number of instances, and F1_i is the F1-Score for class i.

A. *Top Category Classification Results*

As outlined in Table I, a diverse set of models was evaluated to identify the most effective approach for predicting the top-level product category. The models ranged from traditional linear classifiers like Logistic Regression and LinearSVC trained on TF-IDF features, to more complex architectures involving ensemble learning and sentence-transformer-based embeddings.

Among all approaches, the weighted soft-voting ensemble achieved the highest performance, delivering a weighted F1 score of 0.8927. This ensemble combined three complementary classifiers: Logistic Regression trained on TF-IDF vectors, XGBoost using the same features, and a Multilayer Perceptron (MLP) trained on MiniLM sentence embeddings. Each component contributed uniquely Logistic Regression offered robustness and efficiency, XGBoost captured non-linear interactions and handled sparse data well, while the MLP leveraged rich semantic representations provided by the transformer embeddings.

This combination allowed the ensemble to generalize effectively across the varied and noisy textual metadata characteristic of Etsy product listings. Additionally, the model was able to outperform other individual classifiers such as XGBoost with TF-IDF (F1 = 0.8636), Logistic Regression with TF-IDF (F1 = 0.8666), and even MLP with MiniLM embeddings (F1 = 0.8034). The model architecture used in the ensemble is depicted in Fig. 1, and it served as the final submission for the top category prediction task.

TABLE I
PERFORMANCE OF MODELS FOR TOP CATEGORY CLASSIFICATION

Embedding Type	Model	Weighted F1-Score
TF-IDF + MiniLM	**Voting Ensemble (XGB, LR, MLP)**	**0.893**
TF-IDF	Logistic Regression	0.867
TF-IDF	XGBoost	0.864
TF-IDF + MiniLM	LinearSVC + XGBoost	0.860
TF-IDF	LinearSVC	0.856
TF-IDF	RidgeClassifier	0.844
MiniLM	MLP Classifier	0.803
TF-IDF	SGDClassifier	0.777
TF-IDF	MultinomialNB (Baseline)	0.639

B. *Bottom Category Classification Results*

As presented in Table II, the performance of various models for bottom category classification varied significantly depending on the feature representation and learning algorithm used. Among all evaluated models, the Linear Support Vector Classifier (LinearSVC) trained solely on TF-IDF features emerged as the best performer, achieving an F1-score of 0.621. This result highlights the strength of linear models in high-dimensional sparse feature spaces, especially when trained on extensive textual signals such as titles, descriptions, and tags.

In contrast, models leveraging MiniLM embeddings combined with multi-layer perceptrons (MLP) exhibited competitive performance, with the MiniLM + One-Hot Hybrid MLP and MiniLM + Categorical Hybrid MLP achieving F1-scores of 0.537 and 0.534 respectively. These approaches aimed to fuse contextual sentence embeddings with categorical metadata to improve classification precision. An oversampling strategy applied to mitigate class imbalance slightly boosted performance to 0.519, though it still trailed the TF-IDF-based LinearSVC.

Models based solely on GloVe embeddings or dimensionality reduction techniques (such as TF-IDF with SVD) underperformed, with F1-scores falling below 0.40. Incremental training strategies using SGDClassifier with MiniLM or categorical fusion yielded F1-scores around 0.247, indicating limited success in handling the complexity of bottom category granularity with simpler online learning methods.

TABLE II
PERFORMANCE OF MODELS FOR BOTTOM CATEGORY CLASSIFICATION

Embedding Type	Model	Weighted F1-Score
TF-IDF	**LinearSVC**	**0.621**
MiniLM	MiniLM + One-Hot Hybrid MLP	0.537
MiniLM	MiniLM + Categorical Hybrid MLP	0.534
MiniLM	MiniLM + Hybrid MLP with Safe Oversampling	0.519
TF-IDF	Multinomial Naive Bayes	0.503
MiniLM	MLP Classifier	0.491
GloVe	MLP Classifier	0.398
TF-IDF + SVD	Logistic Regression (after dimensionality reduction)	0.340
MiniLM	SGDClassifier with incremental training	0.247
TF-IDF + Categorical	SGDClassifier with class balancing	0.247

C. *Visualization of Learned Representations*

To gain further insight into the semantic patterns learned during training, t-SNE was applied to the MiniLM sentence embeddings of product listings. This helped visualize how well the model could group similar items based on textual metadata such as title, description, and tags.

For the top category classification, the output of the final hidden layer of the fine-tuned MLP was extracted and reduced to two dimensions. As shown in Fig. 3, products belonging to the same high-level categories, such as clothing, shoes, and home and living, form distinct clusters. This clear separation suggests that the model has successfully captured high-level semantic signals across diverse product types.

For the bottom category classification, a similar visualization was performed on MiniLM embeddings of the 50 most frequent bottom-level categories. The results, depicted in Fig. 4, reveal meaningful clustering despite the higher granularity of labels. Many fine-grained categories such as earrings, lamp bases and parts, and diaper caddies form dense clusters, while others with overlapping semantics exhibit closer proximities. Each point represents a product listing in the validation set, color-coded by its top-level category

label.The distinct clustering indicates meaningful separation in the learned embedding space.

Together, these visualizations validate that the embedding space is sufficiently structured to support both coarse-grained (top-level) and fine-grained (bottom-level) classification. Furthermore, they demonstrate that pre-trained embeddings like MiniLM can effectively separate product categories without task-specific fine-tuning, though further fine-tuning may enhance performance even more.

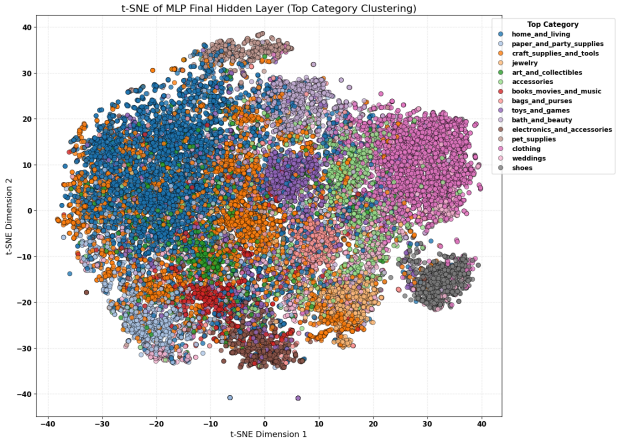

Fig. 3. t-SNE visualization of the final hidden layer of the MLP trained on MiniLM embeddings for top category classification.

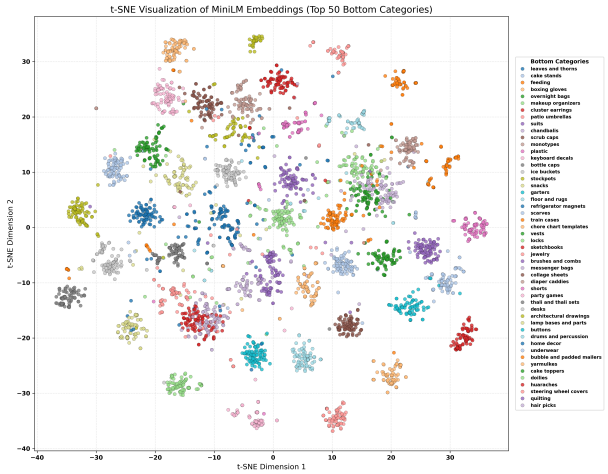

Fig. 4. t-SNE visualization of MiniLM embeddings for the top 50 bottom-level product categories.

D. *Comparison of Pre-trained Embedding Approaches and Task-specific Models*

To explore how general-purpose embeddings perform compared to task-adapted models, several widely used pre-trained embeddings from the literature were evaluated on the top category classification task. These included word-level embeddings such as GloVe and FastText, sentence-level models like the Universal Sentence Encoder (USE), and Word2Vec Skip-gram representations. Each embedding was paired with lightweight classifiers following established approaches. Additionally, Bag-of-Words representations using n-grams were assessed using both LinearSVC and Naive Bayes.

As shown in Table III, the performance of these pre-trained methods varied considerably. While some sentence embeddings performed competitively, traditional n-gram models often outperformed them, especially when paired with strong linear classifiers. However, none of the pre-trained models matched the effectiveness of the fine-tuned ensemble developed in this study.

The final ensemble combining TF-IDF features with XGBoost, logistic regression, and a MiniLM-based MLP demonstrated clear advantages in both accuracy and adaptability. These findings reinforce the benefit of customizing models to the specific domain and dataset rather than relying solely on off-the-shelf embeddings.

This comparison was limited to the top category task, as the best-performing bottom category model was based entirely on TF-IDF features without embedding integration.

TABLE III
COMPARISON OF PRE-TRAINED EMBEDDING MODELS VS. FINE-TUNED ENSEMBLE FOR TOP CATEGORY CLASSIFICATION

Representation Type	Model	Weighted F1-Score
Fine-tuned Hybrid Model	**Voting Ensemble (XGB + LR + MLP)**	**0.893**
Lexical (n-grams)	BoW (1–3g) + LinearSVC [14]	0.818
Lexical (n-grams)	BoW (1–3g) + Multinomial Naive Bayes [14]	0.753
Pre-trained Sentence Embedding	USE + Logistic Regression [15]	0.739
Pre-trained Word Embedding	Word2Vec Skip-gram + LinearSVC [14]	0.706
Pre-trained Word Embedding	FastText + Logistic Regression [16]	0.689

V. CONCLUSION

This study investigated a variety of machine learning approaches for the task of automated product categorization using real-world data from an e-commerce platform. The classification task was tackled at two hierarchical levels, top and bottom categories, leveraging both lexical representations, such as TF-IDF, and semantic embeddings, including pre-trained and hybrid models. The objective was to identify models that balance performance and efficiency when applied to large, noisy datasets typical of online marketplaces.

A. *Challenges*

A key challenge encountered during experimentation was the limitation of computational resources. All models were trained and evaluated on a MacBook Air M2 with 8 GB of RAM, without access to GPU acceleration. This restricted the feasibility of using resource-intensive architectures such as transformer-based models in a fine-tuned setting. As a result, the focus was placed on lightweight models like linear classifiers and shallow neural networks, with sentence embeddings used in frozen form. Emphasis was placed on models that offered fast training times, stability, and memory efficiency without compromising significantly on predictive performance.

B. *Future Work*

Given access to more powerful hardware, future research can explore full fine-tuning of transformer-based language models such as BERT or MiniLM, which were not feasible under current constraints. Hierarchical classification frameworks also present an interesting direction, allowing for a structured approach to model the dependency between top and bottom categories. Furthermore, techniques like contrastive learning could be employed to enhance the discriminative power of learned representations. For bottom category classification in particular, future work may benefit from class-aware loss functions and advanced oversampling strategies to handle the substantial class imbalance more effectively.

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